F12 4th CAR

Lecture 5: Section A.5
Solving Equations

Solving linear equations

Solving rational equations

Extraneous solutions

Solving quadratic equations

Completing the square

Quadratic formula

Discriminant

Solving equations with radicals or rational exponents

Solving absolute value equations

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I need to look at
#5 on prac.

Lecture 5: Section A.5
Solving Equations

x^2-9 expression

Def. An **equation** in x is a statement that two algebraic expressions are equal. $x^2-9=0$

To **solve** an equation is to find all values of x for which the equation is true.

Such values of x are **solutions** (or **roots, zeros**) of the equation.

NOTE: If an equation has no solution, the solution set is **empty**, written as $\emptyset$. or $\{\ \}$ $\cancel{\{0\}}$

ex. Solve the equation $|x-1| = -2$

$\boxed{\emptyset}$

Linear Equations

Def. A **linear equation in one variable** x is an equation of the form

exponent is 1.

$$ax + b = 0$$

where a and b are real numbers with $a \neq 0$.

To solve a linear equation:

1. Remove all parentheses and simplify each side of the equation as much as possible.
2. Rewrite the equation by **isolating the variable**: variable terms on one side, numbers on the other.
3. Solve for the variable and check your solution.

ex. Solve $6(x-1)+4=3(7x+1)$

$6x-6+4 = 21x+3$

$-5 = 15x$

$x = \frac{-5}{15} = \boxed{-\frac{1}{3}}$ can easily check your sol.

NOTE: A linear equation has exactly **one** solution.

Solve a linear equation with fractions, multiply both sides by LCD to clear the fraction.

ex. $4\left(\frac{x}{2}+1\right)=\left(\frac{1}{4}(x-6)\right)4$

$2x+4 = x-6$

$x = -10$

LCD: $(x-2)(x-1)$

To solve rational equations that lead to linear equations

1. Find the domain of the variable. D: $x \neq 2, 1$
2. Clear the equation of fractions by multiplying both sides by LCM of the denominator.
3. Solve for the variable.
4. Choose the solutions that are **in the domain**.

$(x-2)(x-1)$ $(x-2)(x-1)$

ex. Solve: $\left(\frac{3}{x-2}\right) = \left(\frac{1}{x-1} + \frac{7}{x^2-3x+2}\right)$ $(x-2)(x-1)$

$3(x-1) = x-2+7$

$3x-3 = x+5$

$2x = 8$

$\boxed{x=4}$ make sure sol. is in the domain ✓

LCD: $(x+2)(x-2)$ D: $x \neq \pm 2$

$(x+2)(x-2)$ $(x+2)(x-2)$

ex. Solve: $\left(\dfrac{2x}{x^2-4}\right) = \left(\dfrac{4}{x^2-4} - \dfrac{3}{x+2}\right)(x+2)(x-2)$

$2x = 4 - 3(x-2)$

$2x = 4 - 3x + 6$

$5x = 10$

$\cancel{x = 2}$ not in domain so $\emptyset$

NOTE: We call $x = 2$ an extraneous solution.

NOTE: An equation with a single fraction on each side can be cleared of denominators by **cross multiplying**.

D: $x \neq 2, 1$ ← short-cut

ex. $\dfrac{3}{x-2} = \dfrac{1}{x-1}$

$3(x-1) = 1(x-2)$

$3x - 3 = x - 2$

$2x = 1$

$\boxed{x = \frac{1}{2}}$

Quadratic Equations

Def. A **quadratic** or **second-degree equation** is an equation that can be written in the general form

$$ax^2 + bx + c = 0$$

where a, b and c are real numbers and $a \neq 0$.

To solve a quadratic equation:

1. **By Factoring**: Use the Zero-Factor property:

   If $ab = 0$, then So set $= 0$

ex. Solve: $3x^2 + 18x = 21$

$\frac{1}{3}(3x^2 + 18x - 21) = (0)(\frac{1}{3})$

$x^2 + 6x - 7 = 0$

$(x+7)(x-1) = 0$

$x + 7 = 0$ $\quad$ $x - 1 = 0$

$x = -7$ $\quad$ $x = 1$ $\quad$ can check sol.

NOTE: The right side of the equation has to be 0 before factoring.

ex. Solve: $(x+3)(x-4)=8$

$x^2-x-12=8$

$x^2-x-20=0$

$(x-5)(x+4)=0$

$x=5 \quad x=-4$

$x^4=16$

$\sqrt[4]{x^4}=\pm\sqrt[4]{6}$

$x=\pm 2$

2. **By Square Root Principle:**

If $u^2=c$, where $c \geq 0$, then $u=\pm\sqrt{c}$.

ex. Solve: $(x+2)^2=5$

$\sqrt{(x+2)^2}=\pm\sqrt{5}$

$x+2=\pm\sqrt{5}$

$x=-2\pm\sqrt{5}$

$x=-2+\sqrt{5}$

also

$x=-2-\sqrt{5}$

3. **By Completing the Square** of
$ax^2 + bx + c = 0,\ a \neq 0$

1) Make sure $a = 1$. If not, divide each term by a.
2) Move the constant to the right side of the equation.
3) Complete the square by adding the square of one-half of the coefficient of x to each side of the equation.
4) Factor the left side as a perfect square.
5) Solve for x using the Square Root Principle.

$\left[\frac{1}{2}\left(-\frac{4}{3}\right)\right]^2 = \left[-\frac{2}{3}\right]^2 = \frac{4}{9}$

$\frac{\sqrt{10}}{\sqrt{9}}$

<u>ex.</u> Solve by completing the square: $\frac{1}{3}(3x^2 - 4x - 2) = (0)\frac{1}{3}$

$x^2 - \frac{4}{3}x - \frac{2}{3} = 0$

$x^2 - \frac{4}{3}x = \frac{2}{3}$

$x^2 - \frac{4}{3}x + \frac{4}{9} = \frac{2}{3} + \frac{4}{9}$

$\left(x - \frac{2}{3}\right)^2 = \frac{10}{9}$

$\sqrt{\left(x - \frac{2}{3}\right)^2} = \pm\sqrt{\frac{10}{9}}$

$x - \frac{2}{3} = \pm\frac{\sqrt{10}}{3}$

$\boxed{x = \frac{2}{3} \pm \frac{\sqrt{10}}{3}}$

or

$\boxed{x = \frac{2 \pm \sqrt{10}}{3}}$

4. **By Quadratic Formula:** If $ax^2 + bx + c = 0$, $a \neq 0$, then

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Proof. Complete the square.

ex. Solve using the quadratic formula:

reduce 1) $\frac{1}{2}(2x^2 + 8x + 8) = 0\left(\frac{1}{2}\right)$

$x^2 + 4x + 4 = 0$

$A = 1$
$B = 4$
$C = 4$

$$x = \frac{-4 \pm \sqrt{4^2 - 4(1)(4)}}{2(1)}$$

$$x = \frac{-4 \pm \sqrt{0}}{2} = \frac{-4}{2} = \boxed{-2}$$

2) $2x(3 - x) = 3$

Finish for HW

Finish for HW $\frac{3 \pm \sqrt{3}}{2}$ $\boxed{x = \frac{3 \pm \sqrt{3}}{2}}$

3) $x^2 + 2x + 2 = 0$

HW

$x = \frac{-2 \pm \sqrt{-4}}{4}$ so $\boxed{\emptyset}$ Since complex number.

Read

Def. The quantity $b^2 - 4ac$ is the **discriminant** of the quadratic equation.

NOTE: The equation $ax^2 + bx + c = 0$,

1) If $b^2 - 4ac > 0$, then the equation has **two** distinct real number solutions.
2) If $b^2 - 4ac = 0$, then the equation has **one** repeated real number solution.
3) If $b^2 - 4ac < 0$, then the equation has **no** real number solutions.

Practice. Use the discriminant to determine the number of real solutions of the equations

1) $2x^2 + 9x + 1 = 0$

2) $2x^2 + 8x + 8 = 0$

Answer. 1) two distinct real solutions 2) one repeated real solution

$x=6$ only one solution

$x^2=6^2$ now two sol.

Equations with Radicals or Rational Exponents

To find the **real solution** of an equation with radicals:

1. **Simplify** the equation if possible.
2. **Isolate the most complicated radical** on one side of the equation.
3. **Raise** both sides of the equation to the index of the radical to eliminate the radical.
4. **Check** for extraneous solution, since raise both sides to an even power may add extraneous solutions.

ex. Solve: $3+\sqrt{3x+1}=x$

$$\left(\sqrt{3x+1}\right)^2=(x-3)^2$$

$$3x+1=x^2-6x+9$$

$$x^2-9x+8=0$$

$$(x-1)(x-8)=0$$

$$x=1 \quad x=8$$

Ch: $x\neq 1$

$$3+\sqrt{3(1)+1}=1$$

$$3+2\neq 1$$

Ch: $\boxed{x=8}$

$$3+\sqrt{3(8)+1}=8$$

$$3+5=8$$

$$8=8 \checkmark$$

ex. Solve: $(x+3)^{2/3} = 4$

$$\left[(x+3)^{2/3}\right]^3 = (4)^3$$

$$(x+3)^2 = 64$$

$$\sqrt{(x+3)^2} = \pm\sqrt{64}$$

$$x+3 = \pm 8$$

$$x = \pm 8 - 3$$

thus $x = 8 - 3 = 5$

$x = -8 - 3 = -11$

$x = 5$ both
$x = -11$ check

ex. Solve: $\sqrt{3x+1} - \sqrt{x-1} = 2$

$$\left(\sqrt{3x+1}\right)^2 = \left(\sqrt{x-1} + 2\right)^2$$

$$3x+1 = x-1 + 2(2)\sqrt{x-1} + 4$$

$$3x+1 = x+3+4\sqrt{x-1}$$

$$\tfrac{1}{2}(2x-2) = \left(4\sqrt{x-1}\right)\tfrac{1}{2}$$

$$(x-1)^2 = \left(2\sqrt{x-1}\right)^2$$

$$x^2 - 2x + 1 = 4(x-1)$$

$$x^2 - 2x + 1 = 4x - 4$$

$$x^2 - 6x + 5 = 0$$

$$(x-1)(x-5) = 0$$

$$x = 1 \quad x = 5$$

must check sol.

They both check.

$x = 1, 5$

Absolute Value Equations

Recall the definition:

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases}$$

Isolate the | |

ex. Solve each equation:

1) $4 + |3 - x| = 2$

$|3-x| = -2$ $\boxed{\emptyset}$ since -2

2) $|x^2 - 6| = x$

if x^2-6 is pos.

$x^2-6 = x$

$x^2 - x - 6 = 0$

$(x-3)(x+2) = 0$

$\boxed{x=3}$ $\cancel{x=-2}$

Must check sol.

if x^2-6 is neg.

$-(x^2-6) = x$

$-1(-(x^2-6)) = (x)(-1)$

$x^2-6 = -x$

$x^2+x-6=0$

$(x+3)(x-2)=0$

$\cancel{x=-3}$ check sol.

$\boxed{x=2}$

NOTE: If the equation involves rational expressions, radicals, rational exponents or absolute values, you must **check** your solution(s) in the original equation.

Let $u = x+1$

then $u^2 = (x+1)^2$

Sub in: $16u^2 + 8u + 1 = 0$

$(4u+1)(4u+1) = 0$

$4u+1 = 0$

$u = -\frac{1}{4}$

now sub back

$u = x+1$

$-\frac{1}{4} = x+1$

$x = \frac{-5}{4}$

Practice.

1) Solve $16(x+1)^2 + 8(x+1)^1 + 1 = 0$

(Hint: Use substitution $u = x+1$)

2) Solve: $x^3 = 6x^2 - 9x$

3) Solve: $x^6 - 6x^3 + 9 = 0$

4) Solve: $x^3 - 3x^2 - 3x + 9 = 0$

5) Solve: $(3x+1)^{-1/2} + 2(3x+1)^{1/2} = 0$

6) Solve: $(x-1)^{2/3} + (x-1)^{1/3} - 12 = 0$

Answer. 1) $-\frac{5}{4}$ 2) 0, 3 3) $\sqrt[3]{3}$ 4) $-\sqrt{3}, \sqrt{3}, 3$

5) no solution 6) $-63, 28$